# Data Science (Brightlearn) 2026 — Day 1

Data → Truthful information and factual.
→ Collection of facts

Data scientist

→ You look into the past and predict the future.
→ Have the ability to look what happened in the past and what would happen in the future.

Past
↓

Future
↓
Machine learning
AI
↓
Build
Algorithms

## In business there are 2 types of data

1st party data → is the data collect as a business
→ Getting the data by selling

~~2rd pa~~

3rd party data → is the data that you get from someone
→ Doing business based on what you ~~hav~~ heard from someone else.
e.g getting information about weather that it ~~and you~~ would rain ~~heavy~~ heavily and you are selling on the streets, you won't sell on that particular day due to weather.
→ Data used to make decision
→ ~~&~~ It's external data.

Data enrichment
→ When you combine 1st party and 3rd party data

Data format

Data format

Unstructured data
- Data that has no structure
- No standard structure
→ Does not follow a specific structure
e.g music, videos, social media posts, pdf, voice notes, websites, podcasts, pictures

Structured data
- Data that is organized
- Arranged in a consistent manner
- Data stored in a table with rows and columns

Data exists so that we can make decisions

SQL – Structured Query Language
We use SQL to aggregate the data i.e to summarize the data.
We use SQL to do data cleaning.

## Day 2

Database → is where we store data.
Memory card is a form of database
USB

Database Management System → An interface we use to manage our database
e.g phone
→ DMS allows us to delete, access, insert update

Relational database → Is a type of database that is used to store data that is structured.
It means the tables can be related to one another using keys
→ It stores structured data.
Is data in a table
→ It stores data in a table with rows and ~~tab~~ columns

## Day 3

Database Management System → Helps us to access, manage, retrieve the data.

Example: Memory card
AWS
Google drive
Microsoft
Apple

Cloud

Relational database
Online Transactional Processing (OLTP) → It's a database that is used at the store.
→ It allows specific functions
* You can create, read, update, delete
→ Is a relational type of database
→ Is not used for complex analysis

Example

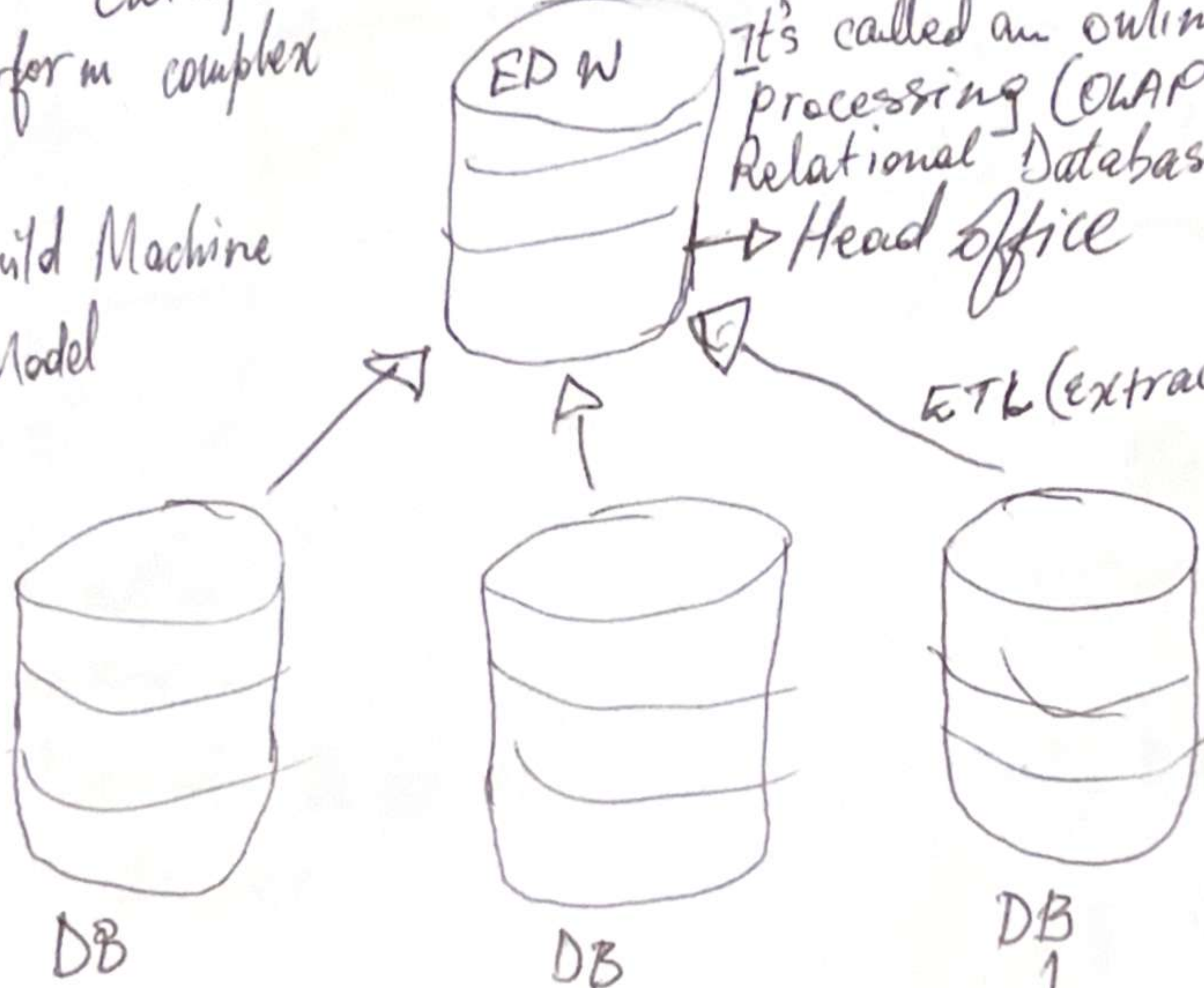
Enterprise DataWarehouse

- You can perform complex analysis
- You can build Machine Learning Model

It's called an online analytical processing (OLAP) Relational Database
→ Head office

ETL (extract Transform Load)

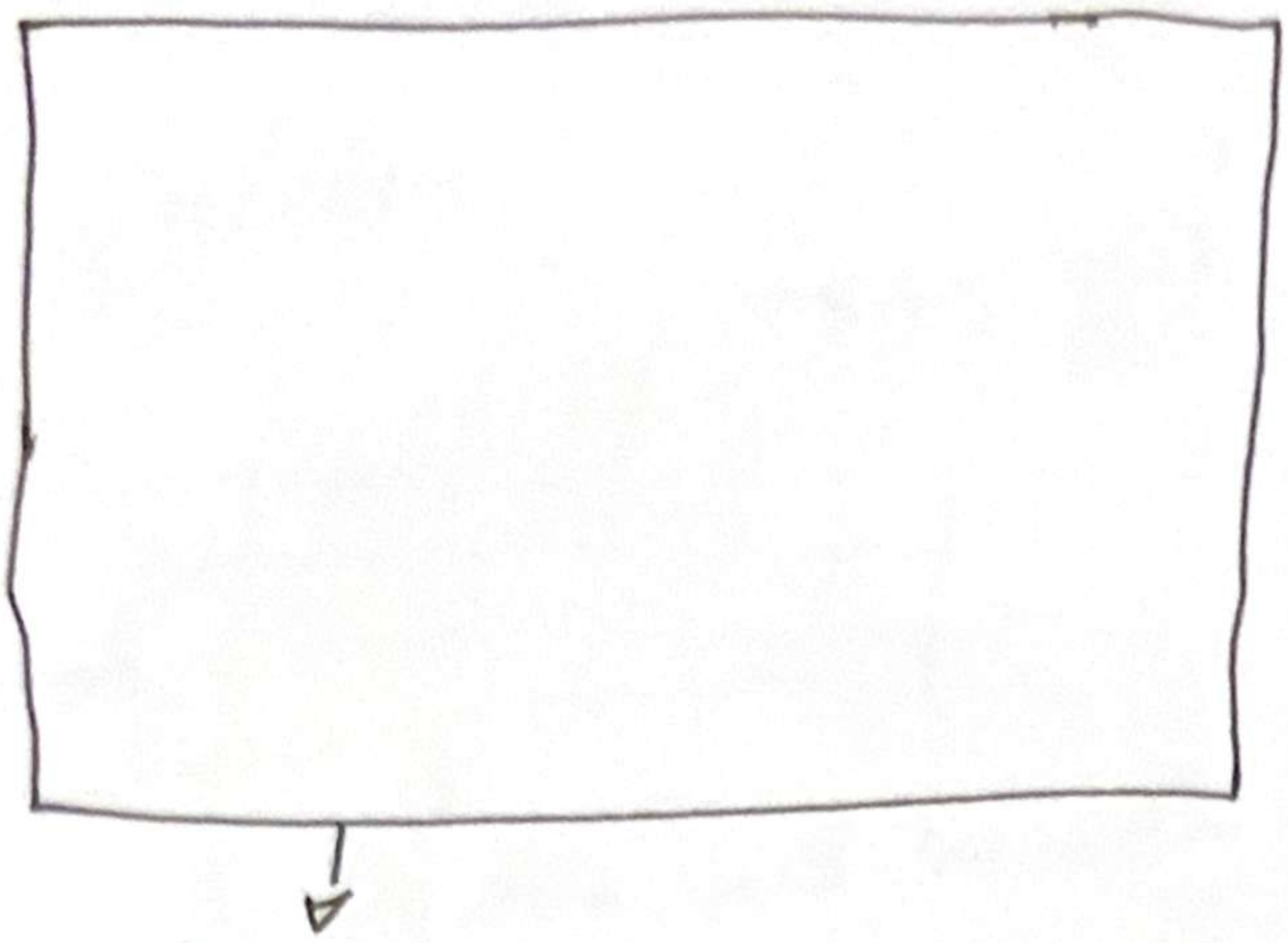
Store



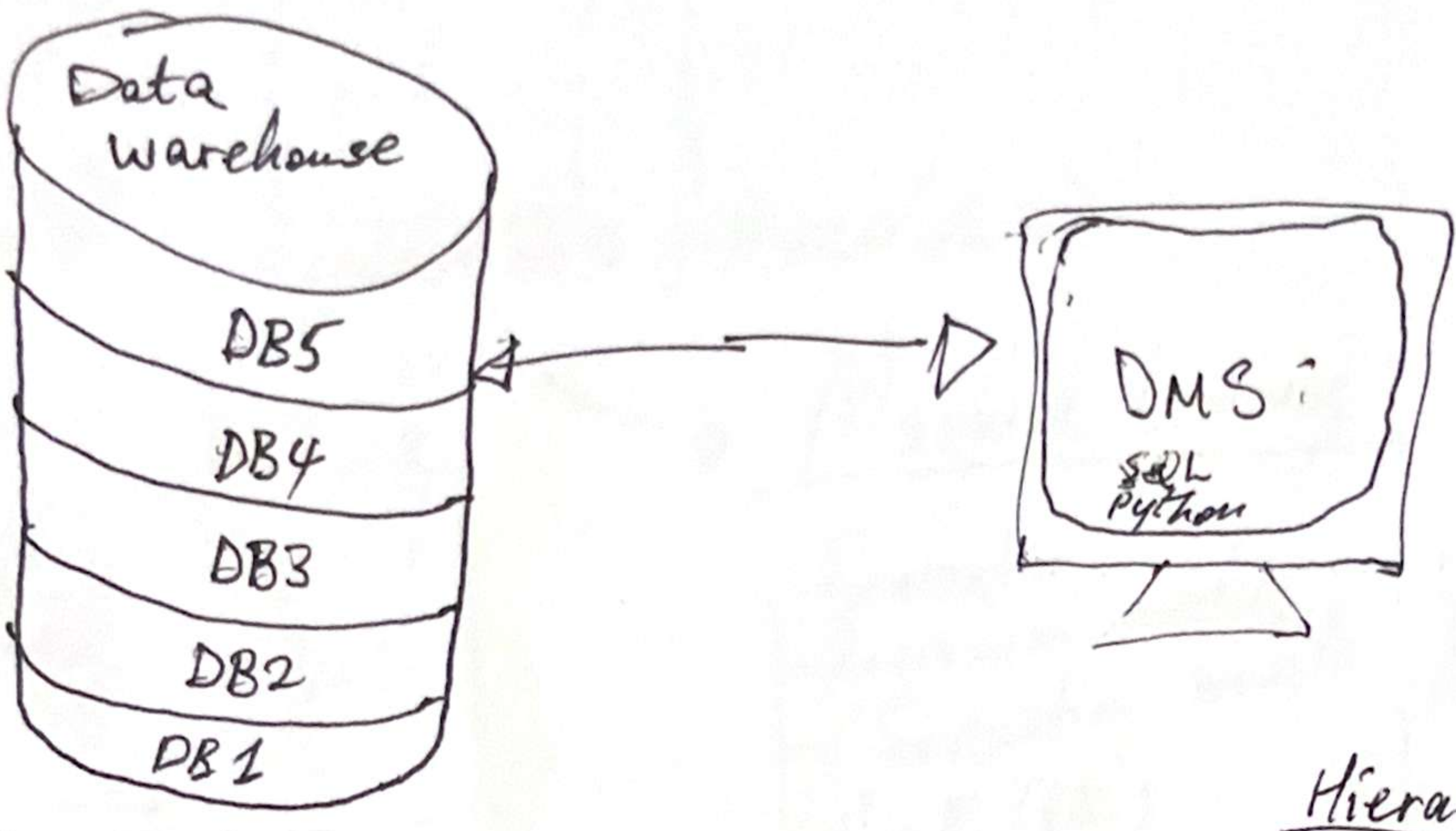

Shoprite branches

4

# Day 4



Data warehouse




Inorder to connect to the data warehouse, you need the Database Management System.

DMS - Snowflake
Databricks
Microsoft SQL Server
DB deaver

## Hierachy

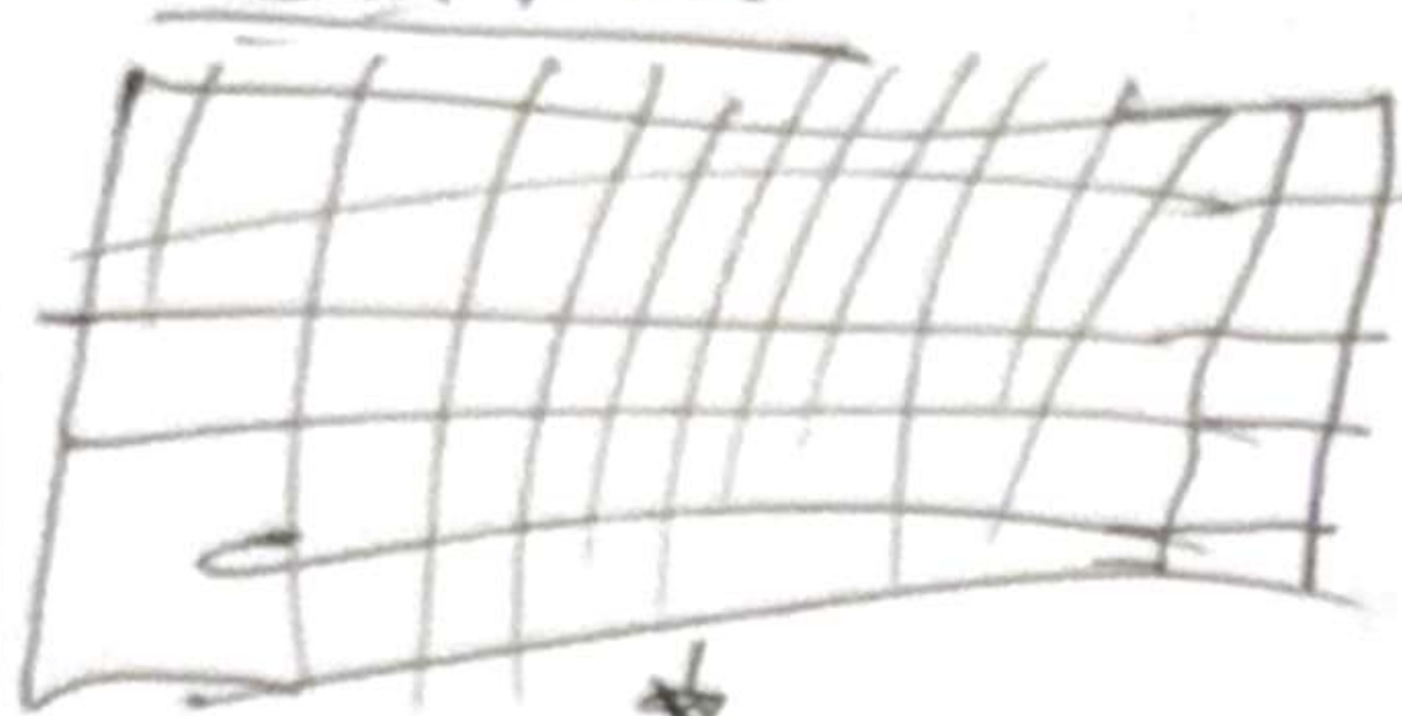
1. Data warehouse
↓
2. Database
↓
3. Schema
↓
4. Tables

Within a database there's a schema

Day 5 SQL Fundamentals

Structured Query Language

## Database



SQL ~~is we use it~~ allows us to clean data, data processing, and data aggregation.

It is a way to summarize the data

Big data with 100 million rows and 100 thousand columns

Summarized data

Aggregated data (Filtered data)

insight

Analytics

Power BI
Excel
Google Looker

Machine Learning

~~Forecasts~~
- Regression models
- Classification models
- GenAI (RAG) (Chatbot)

Features

Examples we give to an algorithm to learn data patterns.

SQL syntax → Is how we write SQL code from the table

SELECT statement → Is used to select specific columns from the table that you want in your final output.

FROM statement → specify the table you are selecting from

database_name.schema_name.table_name

Example: Curro.students.marks

database schema Table name

~~SELECT column_name~~

```
SELECT Column 1,
       Column 2,
       Column 3
FROM database.schema.table;
```

Curro.students.marks

| Student Id | Name | Subject | Marks | Grade |
|---|---|---|---|---|
| 101 | Mbal | Maths | 30% | 9 |
| 102 | Rofhi | Maths | 99% | 12 |
| 103 | Lerato | Geo | 50% | 11 |
| 104 | Thando | Geo | 40% | 11 |

```
SELECT name,
       grade
FROM Curro.students.marks;
```

| Name | Grade |
|---|---|
| Mbali | 9 |
| Rofhi | 12 |

# Day 6 SQL Fundamentals (Filtering)

## SQL Filtering statements

**WHERE** ~~st~~ statement → It is used to filter the rows in the data
→ allows us to select specific rows with specific data.
→ We specify a condition to filter the data.

```
SELECT column1,
       Column 2
       Column 3
FROM database.schema.table
WHERE condition;
```

* We filter the columns using the SELECT statement.
* We filter the rows using the WHERE statement.

SQL Comparison operators → We use the comparison operator in the WHERE clause to create conditions for filtering data.

| Operator | Description |
|---|---|
| = | Equal to |
| != ; # <> | Not equal to |
| > | Greater than |
| >= | Greater than or equal to |
| < | Less than |
| <= | Less than or equal to |

WHERE affects the rows in our data.

curro.student.marks

| Student id | Name | Subject | Marks |
|---|---|---|---|
| 101 | Rofhi | Maths | 99 |
| 102 | Mbali | Maths | 22 |
| 103 | Fifi | Geo | 50 |
| 104 | Tumelo | Bio | 40 |

```
SELECT Name, marks
FROM curro.student.mark
WHERE marks < 50;
```

| Name | Marks |
|---|---|
| Mbali | 22 |
| Tumelo | 40 |

```
SELECT name, subject, marks
FROM curro.students.marks
WHERE marks > 50;
```

| Name | Subject | Marks |
|---|---|---|
| Rofhi | Maths | 99 |

Day 7

**OR statement** → allows us to specify multiple conditions but atleast one condition must be met.

```
SELECT column1,
       column2,
       column3
FROM database.schema.table
WHERE condition1 OR condition2;
```

SELECT *<br>
FROM curro.students.marks<br>
WHERE marks > 50 OR subject = ~~'Acc'~~ ~~'Geo'~~ 'Acc';

| Student_id | Name | Subject | Marks |
|---|---|---|---|
| 101 | Rofhi | Maths | 99 |
| 103 | Mbali | Geo | 50 |
| 104 | Ally | Acc | 40 |

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Day 8

**IN statement** → Allows us to specify multiple values to choose from when filtering.

```
SELECT column1,
       column2,
       column3
FROM table
WHERE column IN(value1, value2, value3)
```

| Student_id | Name | Subject | Marks |
|---|---|---|---|
| 101 | Rofhi | Maths | 99 |
| 102 | Mbali | Maths | 22 |
| 103 | Lerato | Geo | 50 |
| 104 | Ally | Acc | 40 |
| 105 | Niko | Science | 70 |

I want the names of students doing Geo and accounting.

```
SELECT name,
       subject
FROM curro.students.marks
WHERE subject IN('Geo', 'Acc');
```

| Name | Subject |
|---|---|
| Lerato | Geo |
| Ally | Acc |

Day8

BETWEEN statement → allows us to specify a range of values in a column.
→ It's inclusive

```
SELECT column1,
       column 2
FROM table
WHERE column BETWEEN min value AND max value;
```

```
SELECT Name,
       subject,
       Marks
FROM curro.student.marks
WHERE marks BETWEEN 0 AND 30;
```

| Name | Subject | Marks |
|---|---|---|
| Mbali | Maths | 22 |

```
SELECT name,
       subject,
       marks
FROM curro.students.marks
WHERE marks BETWEEN 50 AND 100;
```

These 2 values are included

| Name | Subject | Marks |
|---|---|---|
| Niko | Science | 70 |
| Rofhi | Maths | 99 |
| Lerato | Geo | 50 |

NOT statement → It gives you the output that you are not looking for.
→ It gives us the ability to exclude values in a column.

```
SELECT column1,
       column 2
FROM database.schema.table
WHERE NOT condition?
```

```
SELECT column1,
       column 2
FROM database.schema.table
WHERE column NOT BETWEEN min value AND max value;
```

```
SELECT column1,
       column 2
FROM database.schema.table
WHERE column NOT IN (value1, value2, value3);
```

```
SELECT name,
       subject,
       marks
FROM curro.students.marks
WHERE marks NOT BETWEEN 50 AND 100;
```

| Name | Subject | Marks |
|---|---|---|
| Mbali | Maths | 22 |
| Ally | Acc | 40 |

```
SELECT Name,
       subject.
FROM curro.students.marks
WHERE subject NOT IN ('Geo', 'Acc');
```

| Name | subject |
|---|---|
| Rofhi | MATHS |
| Mbali | Maths |
| Niko | science |